Integration

Week 5C

Overview

Integration of other Functions

Key Ideas - Integration Logarithmic Functions

- Integration Trigonometric Functions

Key Idea - Integration of Logarithmic Functions

Again, we remind ourselves about the result of which states that integration and differentiation are inverse processes.

$$\frac{d}{dx}F(x)=f(x)\Leftrightarrow\int f(x)\,dx \;=F(x)+c$$

So,

$$\text{If } \frac{d}{dx}\ln x=\frac{1}{x} \text{ and } \frac{d}{dx}\ln f(x)=\frac{f'(x)}{f(x)}$$

It follows that,

$$\int\frac{1}{x}\,dx \;=\ln|x|+c \text{ and } \int\frac{f'(x)}{f(x)}\,dx \;=\ln|f(x)|+c$$

Examples:

Integrate the following:

(a) $$\int \frac{2x}{x^2+1}\,dx = \ln\left|(x^2+1)\right| + c$$

(b) $$\int \frac{x^2}{x^3+1}\,dx = \frac{1}{3}\int \frac{3x^2}{x^3+1}\,dx$$
$$= \frac{1}{3}\ln(x^3+1) + c$$

(c)

$$\int \frac{1}{5x}\,dx = \frac{1}{5}\int \frac{1}{x}\,dx$$
$$= \frac{1}{5}\ln|x| + c$$

alternatively

$$\int \frac{1}{5x}\,dx = \frac{1}{5}\int \frac{5}{5x}\,dx$$
$$= \frac{1}{5}\ln|x| + c$$

(d) $$\int \frac{6x}{x^2+1}\,dx = \int \frac{3(2x)}{x^2+1}\,dx$$
$$= 3\int \frac{2x}{x^2+1}\,dx$$
$$= 3\ln\left|x^2+1\right|$$

alternatively

$$\int \frac{6x}{x^2+1}\,dx = 6\int \frac{x}{x^2+1}\,dx$$
$$= 6\times\frac{1}{2}\int \frac{2x}{x^2+1}\,dx$$
$$= 3\ln\left|x^2+1\right|$$

Use $\int \frac{f'(x)}{f(x)}\,dx = \ln|f(x)| + C$

Exercise:

Integrate the following:

(a) $\int \frac{2x}{x^2+7}\,dx = \ln|x^2+7| + C$

$f(x) = x^2+7$

$f'(x) = 2x$

(b) $\int \frac{2x+7}{x^2+7x-3}\,dx = \ln|x^2+7x-3| + C$

(c) $\int \frac{x}{x^2-3}\,dx$ $= \frac{1}{2}\int \frac{2x}{x^2-3}$ $= \frac{1}{2}\ln\left|x^2-3\right| + c$

$f(x) = x^2-3$

$f'(x) = 2x$

(d) $\int \frac{x-2x^3}{x^2-x^4}\,dx$ $= \frac{1}{2}\int \frac{2(x-2x^3)}{x^2-x^4}$

$f(x) = x^2 - x^4$

$f'(x) = 2x - 4x^3$

$= \frac{1}{2}\ln\left|x^2-x^4\right| + c$

Exercise:

Find the value of $\int_2^4 x - \frac{1}{x}\,dx$.

$$= \left[\frac{x^2}{2} - \ln|x|\right]_2^4$$

$$= \left(\frac{4^2}{2} - \ln 4\right) - \left(\frac{2^2}{2} - \ln 2\right)$$

$$= 8 - \ln 4 - 2 + \ln 2$$

$$= 6 + \ln 2 - \ln 4$$

$$= 6 + \ln\left(\frac{2}{4}\right)$$

$$= 6 + \ln\left(\frac{1}{2}\right)$$

$$= 6 + \ln 1 - \ln 2$$

$$= 6 + 0 - \ln 2$$

Key Idea - Integration of Trigonometric Functions

If , $\frac{d}{dx}\sin x = \cos x$ and $\frac{d}{dx}\sin f(x) = \cos f(x) + c$

Therefore, it follows that,

$$\int \cos x \, dx = \sin x + c \quad \text{and} \quad \int f'(x)\cos f(x) \, dx = \sin f(x) + c$$

If , $\frac{d}{dx}\cos x = -\sin x$ and $\frac{d}{dx}\cos f(x) = -\sin f(x) + c$

Therefore, it follows that,

$$\int \sin x \, dx = -\cos x + c \quad \text{and} \quad \int f'(x)\sin f(x) \, dx = -\cos f(x) + c$$

If, $\frac{d}{dx}\tan x = \sec^2 x$ and $\frac{d}{dx}\tan f(x) = \sec^2 f(x) + c$

Therefore, it follows that,

$$\int \sec^2 x \, dx = \tan x + c \quad \text{and} \quad \int f'(x)\sec^2 f(x) \, dx = \tan f(x) + c$$

Examples: Evaluate the following integrals:

(a) $\int 5\sin(5x+1)\,dx = -\cos(5x+1)+c$

(b) $\int \sin 3x\,dx = \frac{1}{3}\int 3\sin 3x\,dx$

$= -\frac{1}{3}\cos 3x + c$

(c) $\int 6x\sec^2(3x^2+1) + x\,dx = \tan(3x^2+1) + \frac{x^2}{2} + c$

(d) $\int \sin\left(\frac{x}{4}\right)dx = 4\int \frac{1}{4}\sin\left(\frac{x}{4}\right)\,dx$

$= -4\cos\left(\frac{x}{4}\right) + c$

(e) $\int 6\sin \cancel{2}^{3}x\,dx = 6\int \sin \cancel{2}^{3}x\,dx$

$= 6\times\frac{1}{3}\int 3\sin 3x\,dx$

$= -2\cos 3x + c$

(f) $\int (x + 8\sin 2x)\,dx = \int x\,dx + \int 8\sin 2x\,dx$

$= \int x\,dx + 8\times\frac{1}{2}\int 2\sin 2x\,dx$

$= \frac{x^2}{2} - 4\cos 2x + C$

Exercise:

Find following integrals:

(a) $\int (\cos x + 6)\ dx$ $=$ $\sin x + 6x + c$

(b) $\int \left(\sec^2 x - 3x\right)\ dx$ $= \tan x - \frac{3x^2}{2} + c$

Exercise:

Find following integrals:

(a) $\int 3\sin(3x-\pi)\ dx$ $= -\cos(3x-\pi)+c$

$f(x) = 3x-\pi$

$f'(x) = 3$

(b) $\int \sec^2(3x+2)\ dx$ $= \frac{1}{3}\int 3 \sec^2(3x+2)$

$= \frac{1}{2}\tan(3x+2)+c$

Exercise:

Find following integrals:

(a) $\int \cos\left(\frac{\pi}{2} - x\right)\, dx$

$$= (-1)\int - \cos\left(\frac{\pi}{2} - x\right)$$

$$= -\sin\left(\frac{\pi}{2} - x\right) + c$$

(b) $\int 5 \sin \frac{x}{2}\, dx$

$$= 5 \times 2 \int \frac{1}{2} \sin\left(\frac{x}{2}\right) dx$$

$$= 10 \int \frac{1}{2} \sin\left(\frac{x}{2}\right) dx$$

$$= -10 \cos\left(\frac{x}{2}\right) + c$$

Exercise:

Find following integrals:

(a) $\int x^2 + 6\cos(4x-3)\,dx$ $= \frac{x^3}{3} + \frac{3}{2}\sin(4x-3) + c$

(b) $\int \frac{\cos 2x}{2}\,dx$ $= \frac{1}{2} \times \cancel{\frac{1}{2}} \int \cancel{2} \cos 2x \; dx$

$= \frac{1}{4} \sin 2x + c$

Exercise:

Find the value of the following:

(a) $\int_0^{\frac{\pi}{2}} \sin x \, dx$

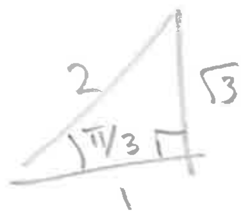
$= \left[-\cos x \right]_0^{\pi/2}$

$= \left(-\cos \frac{\pi}{2}\right) - \left(-\cos 0\right)$

$= 0 - (-1)$

$= 0 + 1$

$= 0$



(b) $\int_{\frac{\pi}{6}}^{\frac{\pi}{2}} \cos(2x) \, dx$

$= \left[\frac{1}{2} \sin(2x) \right]_{\pi/6}^{\pi/2}$

$= \frac{1}{2} \left[\sin 2x \right]_{\pi/6}^{\pi/2}$

$= \frac{1}{2} \left[\sin 2 \times \frac{\pi}{2} - \sin 2 \times \frac{\pi}{6} \right]$

$\frac{1}{2} \left[\sin \pi - \sin \frac{\pi}{3} \right]$

$= \frac{1}{2} \left[0 - \frac{\sqrt{3}}{2} \right]$

$= \frac{-\sqrt{3}}{4}$

Mixed Integration

This final part of the lecture brings together all the integration techniques we have covered so far. The aim is to practise recognising the structure of the integrand and selecting the most appropriate method.

Exercise:

Integrate the following:

(a) $\int \frac{1}{x^8} + \sin(2x)\ dx$

$$= \int x^{-8} + \sin(2x)\,dx$$

$$= \frac{x^{-7}}{-7} - \frac{1}{2}\cos 2x + c$$

$$= \frac{-1}{7x^7} - \frac{\cos 2x}{2} + c$$

(b) $\int \sec^2 4x + \sqrt{x}\ dx$

$$= \int \sec^2 4x + x^{\frac{1}{2}}\ dx$$

$$= \frac{1}{4}\tan 4x + \frac{2x^{3/2}}{3} + c$$

$$= \frac{\tan 4x}{4} + \frac{2\sqrt{x^3}}{3} + c$$

(c) $\int xe^{2x^2+3} + \sec^2(3x-1)\,dx$

$= \frac{1}{4}e^{2x^2+3} + \frac{1}{3}\tan(3x-1) + c$

(d) $\int \frac{2}{3x+9} - \sin\left(\frac{x}{4}\right)\, dx$

$$= \frac{2}{3} \ln |3x+9| + 4\cos\left(\frac{x}{4}\right) + C$$